

Glen A. Dsouza

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Engineering Portfolio: [flame1113.github.io/Portfolio](https://github.com/Flame1113/Portfolio) — full write-ups, 3D models, schematics & build logs for every project below

EDUCATION

California Polytechnic State University, San Luis Obispo

Expected May 2030

B.S. Electrical Engineering — College of Engineering

TECHNICAL PROJECTS

Power Wheels — Centralized Vehicle Control Unit (CVCU)

June 2026 – August 2026

Designer & Integrator

- Designed a custom PCB as the centralized control hub for a modified Power Wheels ride-on, integrating an ESP32-WROOM-32U for telemetry, RFID access control, an OLED UI, and TPA3118-based audio amplification.
- Implemented AO3400 N-channel MOSFET propulsion switching with SPDT relay isolation and 30A motor terminals; populated the board via reflow and precision hot-plate soldering.
- Programmed WS2815 addressable LED lighting effects and built a self-hosted, mobile-responsive web UI on the ESP32 for real-time dashboard control of lighting, audio, and RFID status over Wi-Fi.
- Diagnosed and resolved recurring SPI/serial faults traced to flux contamination and unshielded signal runs, improving RFID and communication reliability.

Custom Mini-Z 1:28 RC Platform

Sep 2025 – Dec 2025

Designer & Fabricator

- 3D-printed the chassis for a 1:28-scale RC platform paired with a custom KiCad controller PCB (ATtiny85, DRV8833PWP) designed under sub-millimeter tolerance constraints.
- Integrated propulsion and steering servo subsystems by reverse-engineering the radio receiver's PWM signal output and deriving timing-to-motion conversion math to calibrate throttle and steering response; independently assembled, soldered, and validated electromechanical performance.

Facial Recognition Operated Door Lock

June 2023 – July 2024

Project Developer

- Built an ESP32-based access control system with a helix keypad, 5V servo actuation, LCD feedback, and buzzer status signaling to enable remote and password-based entry.
- Developed firmware in Arduino C++ for password validation and lock-state management, and integrated an ESP32-CAM with a Python/OpenCV facial recognition pipeline for camera-based access.

LEADERSHIP

Key Club (Kiwanis International)

Club President

- Grew club membership from 5 to over 30 members, coordinating officer teams and planning service events.
- Connected with other Key Clubs across the division to collaborate on joint service initiatives and best practices.
- Presented on core Key Club principles and programs to the full division of 200+ members and officers.

EXPERIENCE

Kumon — Thousand Oaks, CA

May 2022 – Dec 2025

Paid Tutor

- Evaluated and graded math and reading assessments for students across grade levels K–12.
- Delivered targeted instructional support to students requiring additional concept reinforcement.
- Opened and independently managed the tutoring center in the owner's absence, overseeing daily operations and maintaining a clean, organized learning environment.

SparkPrintz — Remote

June 2024 – Aug 2024

Sales & Marketing Intern

- Cold-emailed school administrators and faculty to market apparel-printing services for an early-stage startup.
- Conducted market research and authored business analysis write-ups on company positioning.
- Participated in professional development panels and discussions with Ivy League graduates, gaining insight into career planning and industry pathways.

TECHNICAL SKILLS

Programming: C++, Python, Java, JavaScript, HTML/CSS, SQL

EE / Design: KiCad, Altium, AutoCAD, PCB Layout, SMD Assembly, Power Electronics

Embedded: ESP32, ATtiny85, Arduino, PWM Motor Control, I²C, UART

CAD / Fab: Fusion 360, FDM 3D Printing, Reflow Soldering, Enclosure Design